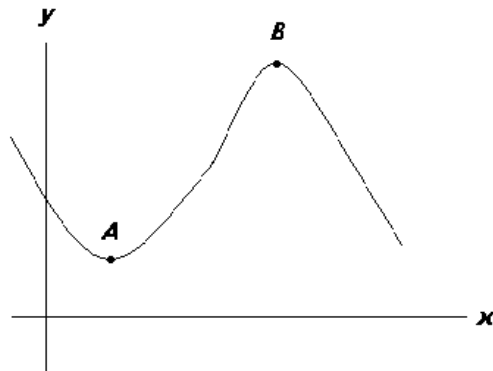


Homework Section 13.3

1. Find the exact value of the length of the curve given by $\mathbf{r}(t) = 2t\mathbf{i} + \cos t\mathbf{j} - \sin t\mathbf{k}$, $-2 \leq t \leq 8$.
2. Set up the integral that gives the length of the following curve and then approximate that length, accurate to the nearest thousandth, using a calculator:
 $\mathbf{r}(t) = 2t\mathbf{i} + e^t\mathbf{j} - 3t^2\mathbf{k}$, $-5 \leq t \leq 6$.
3. Find the arc length function (starting from the point corresponding to $t = 0$) and use it to reparameterize $\mathbf{r}(t)$ in terms of arc length.
 - a) $\mathbf{r}(t) = 5t\mathbf{i} + \cos t\mathbf{j} + \sin t\mathbf{k}$
 - b) $\mathbf{r}(t) = \langle 3t, 1 + 2t, 4 - t \rangle$
4. Find the unit tangent and unit normal vectors:
 - a) $\mathbf{r}(t) = 2t\mathbf{i} + \cos t\mathbf{j} - \sin t\mathbf{k}$
 - b) $\mathbf{r}(t) = e^{-t}\mathbf{i} + e^t\mathbf{j} - \sqrt{2}t\mathbf{k}$
5. Find the curvature:
 - a) $\mathbf{r}(t) = 2t\mathbf{i} + \cos t\mathbf{j} - \sin t\mathbf{k}$
 - b) $\mathbf{r}(t) = e^{-t}\mathbf{i} + e^t\mathbf{j} - \sqrt{2}t\mathbf{k}$
6. Find the curvature: $\mathbf{r}(t) = 2t\mathbf{i} + (1 - t^2)\mathbf{j} - 3t\mathbf{k}$.
7. Find the curvature of $\mathbf{r}(t) = 2t\mathbf{i} + (1 - t^2)\mathbf{j} - 3t\mathbf{k}$ at the point $(2, 0, -3)$.
8. Find the curvature of $y = 8x^4 - x$.
9.
 - a) Where does the below curve have greater curvature, point A or point B? Justify your answer.
 - b) Sketch the Circle of Curvature at point A and use its radius to approximate the curvature of the given curve at point A. Use millimeters as your units.



10. Find the unit tangent, principal unit normal, and binormal vectors of $\mathbf{r}(t) = \frac{1}{3}t^3\mathbf{i} - \frac{\sqrt{2}}{2}t^2\mathbf{j} - t\mathbf{k}$ at the point given by $t = 1$.
11. Find the equations of the normal plane and osculating plane of $\mathbf{r}(t) = \frac{1}{3}t^3\mathbf{i} - \frac{\sqrt{2}}{2}t^2\mathbf{j} - t\mathbf{k}$ at the point given by $t = 1$.
12. Let $\mathbf{r}(t)$ graph to be a line in space (i.e. $\mathbf{r}(t) = \langle x_1 + at, y_1 + bt, z_1 + ct \rangle$). Find the principal unit normal vector, $\mathbf{N}(t)$, in relation to $\mathbf{r}(t)$, if it exists.